

Transport Management System

A Java Inheritance Exercise — if-else, for loops, variables & methods

Scenario Overview

Build a Java program that models different types of transport using **inheritance**. A parent class **Transport** defines shared behavior, while child classes **Bus** and **Bike** extend it, override a method, and add their own specific behavior using **if-else** conditions and a **for loop**.

What This Exercise Tests

Concept	Where You Use It
Inheritance	Bus extends Transport, Bike extends Transport
Method overriding	showWheels()
if-else	showSpeed(), passengers(), checkHelmet()
for loop	printPassengers()
Method parameters	showSpeed(int speed)
Boolean	checkHelmet(boolean helmet)
Object creation	new Bus(), new Bike()

Parent Class: Transport

Create a class **Transport** with the following methods:

- **start()** — prints "Transport started"
- **showSpeed(int speed)** — prints "High Speed" if speed > 80, otherwise prints "Normal Speed"
- **showWheels()** — prints "Transport has wheels"

Child Class: Bus extends Transport

1. Override showWheels()

Print: "Bus has 6 wheels"

2. Method: passengers(int count)

If passenger count > 40 → print "Bus is almost full", otherwise → print "Seats are available"

3. Method: printPassengers(int count)

Use a **for loop** to print passenger numbers. For example, if count = 5:

```
Passenger 1  
Passenger 2  
Passenger 3  
Passenger 4  
Passenger 5
```

Child Class: **Bike** extends Transport

1. Override showWheels()

Print: "Bike has 2 wheels"

2. Method: checkHelmet(boolean helmet)

If helmet is **true** → print "Helmet is worn", otherwise → print "Please wear helmet"

Main Class

Create objects and call the following methods:

```
Bus b = new Bus();  
Bike bike = new Bike();
```

For the Bus object, call:

```
start()  
showWheels()  
showSpeed(70)  
passengers(35)  
printPassengers(5)
```

For the Bike object, call:

```
start()  
showWheels()  
showSpeed(100)  
checkHelmet(false)
```

Expected Output

Your program should produce output similar to:

```
Transport started
Bus has 6 wheels
Normal Speed
Seats are available
Passenger 1
Passenger 2
Passenger 3
Passenger 4
Passenger 5
Transport started
Bike has 2 wheels
High Speed
Please wear helmet
```

TIP — Try writing the code yourself first. This is a good next exercise after your Vehicle → Car / Bike / Tractor example because the inheritance structure is similar, but now you have to combine it with control statements (if-else) and loops (for).

Class Structure at a Glance

Transport (Parent)	Bus (Child)	Bike (Child)
start()	showWheels() [override]	showWheels() [override]
showSpeed(int speed)	passengers(int count)	checkHelmet(boolean helmet)
showWheels()	printPassengers(int count)	